

Custom Handheld Game Console

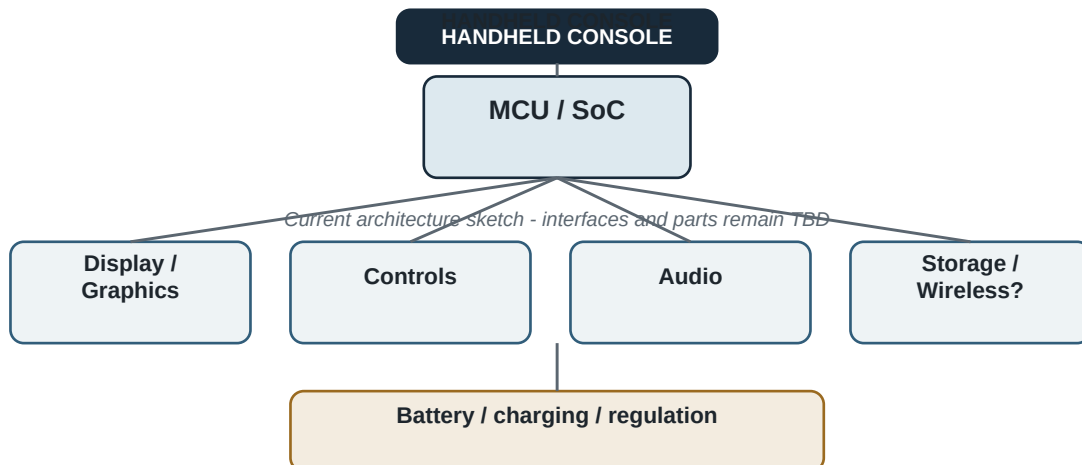
Month 1 Project Update | CMPE Senior Project | August-September 2026

Month 1 was about turning a broad senior-project idea into an engineering direction. The project is currently taking shape as a custom handheld game console, with the team deliberately keeping major hardware choices open until the requirements are clearer.

Where the project started

The project did not begin as a game console. The team initially explored an FPGA/hardware-accelerator direction, then pivoted toward a system that could still demonstrate meaningful hardware and embedded engineering while producing a complete, tangible device. The current concept is a small custom game console built around a microcontroller or high-performance MCU/SoC, with C/C++ firmware, a graphical display, physical controls, audio, and eventually a custom PCB and enclosure.

The handheld form factor is now the working direction, but the team is still treating many implementation details as open engineering questions rather than fixed requirements.



What we explored this month

Area	Current direction	Status
Processor / platform	Compared ESP32-P4 and several STM32 families; Raspberry Pi Pico was also mentioned earlier.	Open
Display / graphics	A graphical display is a core requirement. Display interface, resolution, framebuffer strategy, and graphics architecture still need to be selected.	Open
Audio	An earlier LM386 amplifier project may serve as a proof-of-concept or reference for the console audio subsystem.	Investigating
Power	Battery operation is expected for the handheld. A custom power subsystem using dedicated regulator/charger ICs is an idea under consideration.	Idea only

Area	Current direction	Status
Mechanical	A 3D-printed enclosure is a likely path for prototypes and the final handheld form.	Early direction

Platform research: no winner yet

On September 4, the team held a short exploratory discussion about processor and development-board options. ESP32-P4 and STM32 platforms were the main focus, with STM32H743, H747, H7R7/H7S7, and N657 families entering the discussion. The purpose was to expand the design space and understand what should be compared; no processor or development board was selected.

Rather than choosing by clock speed alone, the team is beginning to look at the system-level factors that matter for a console: display support, graphics acceleration, RAM and external memory, audio peripherals, storage, wireless, power consumption, development tools, cost, and how difficult the final custom PCB would be.

Reusing earlier work: audio

One useful connection to earlier coursework is an existing LM386 audio-amplifier PCB. The original project progressed from LTspice simulation through KiCad schematic capture, PCB layout, assembly, and speaker testing. Rather than automatically reusing that exact board, the current idea is to treat it as a reference or early proof-of-concept for the console audio chain. A console-specific circuit could later be redesigned around the actual speaker, supply, gain, filtering, size, and power requirements.

A possible power-electronics direction

Because the console is trending toward a battery-powered handheld, power management may become a substantial part of the hardware design. One idea being explored is a middle-ground approach: use proven charger and switching-regulator ICs, but design the surrounding power circuitry, component selection, PCB layout, and validation in-house. No decision has been made; the sensible time to choose the power architecture is after the display, processor, audio, and real current requirements are better understood.

Where things stand after Month 1

The biggest result of the first month is not a selected chip or finished prototype. It is a clearer design space: a handheld console with graphics, controls, audio, battery power, and an eventual custom PCB - with the major implementation choices still open.

Current working goals

- **Define the minimum viable console.** Agree on the core features needed for a successful senior project before optimizing stretch goals.
- **Select the development platform.** Compare the strongest MCU/SoC candidates against actual display, memory, audio, wireless, power, and custom-PCB requirements.
- **Bring up display and audio in parallel.** Once the platform is chosen, split early prototyping so one pair can focus on graphics/display while another investigates audio.

- **Prototype first, integrate later.** Use development boards and existing hardware to prove the architecture before committing to the final PCB.
- **Measure before designing the final power system.** Let real subsystem current and voltage requirements drive the battery and regulator architecture.

What I want to show next

The next meaningful update should move from research into hardware bring-up: a selected development platform, a display producing controlled graphics, an MCU-generated sound path, and an initial system-level architecture that can be tested rather than only discussed.

Project status: Early architecture and component research. No final MCU, display, audio architecture, or power architecture has been selected as of September 11, 2026.